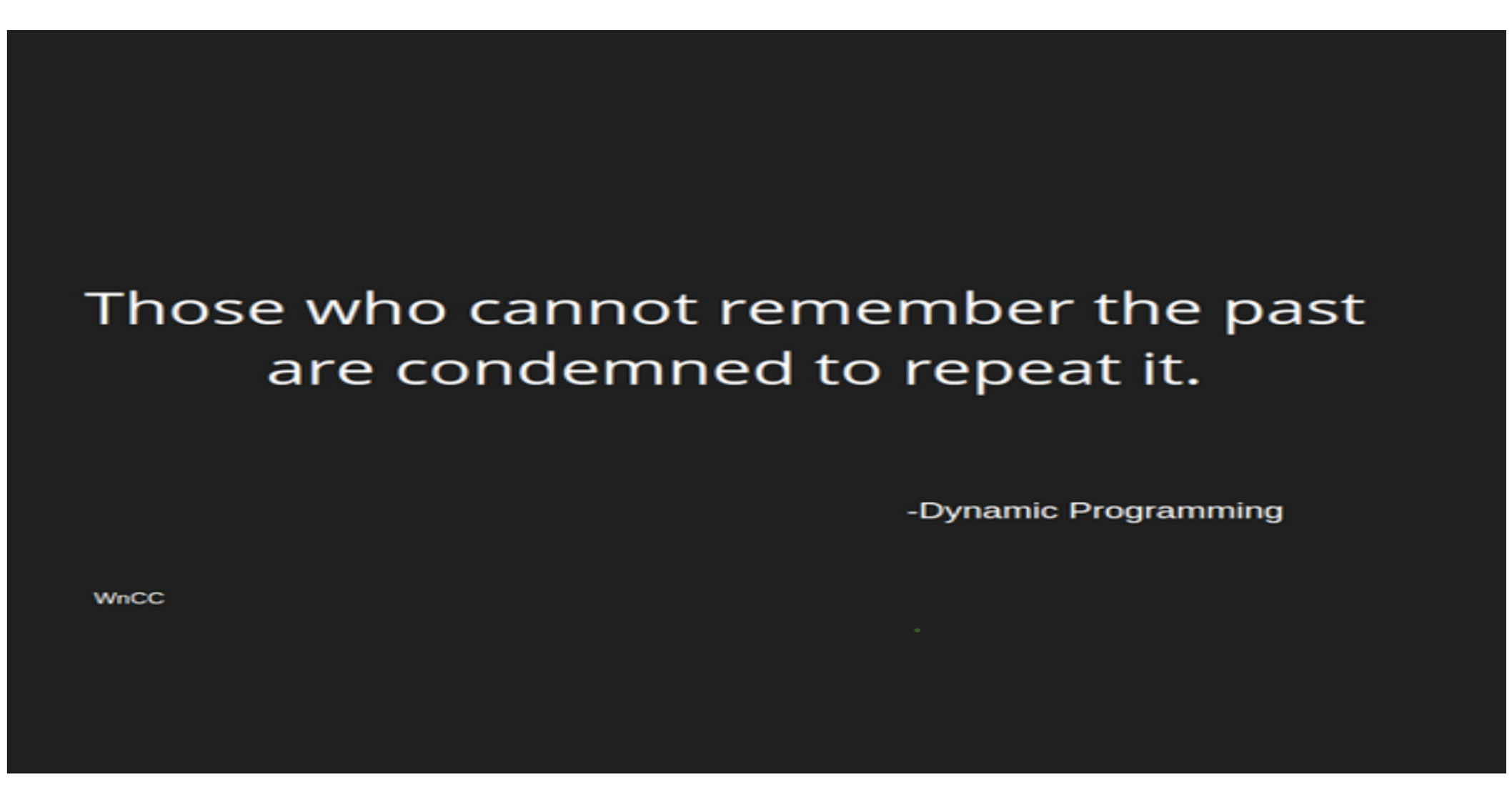
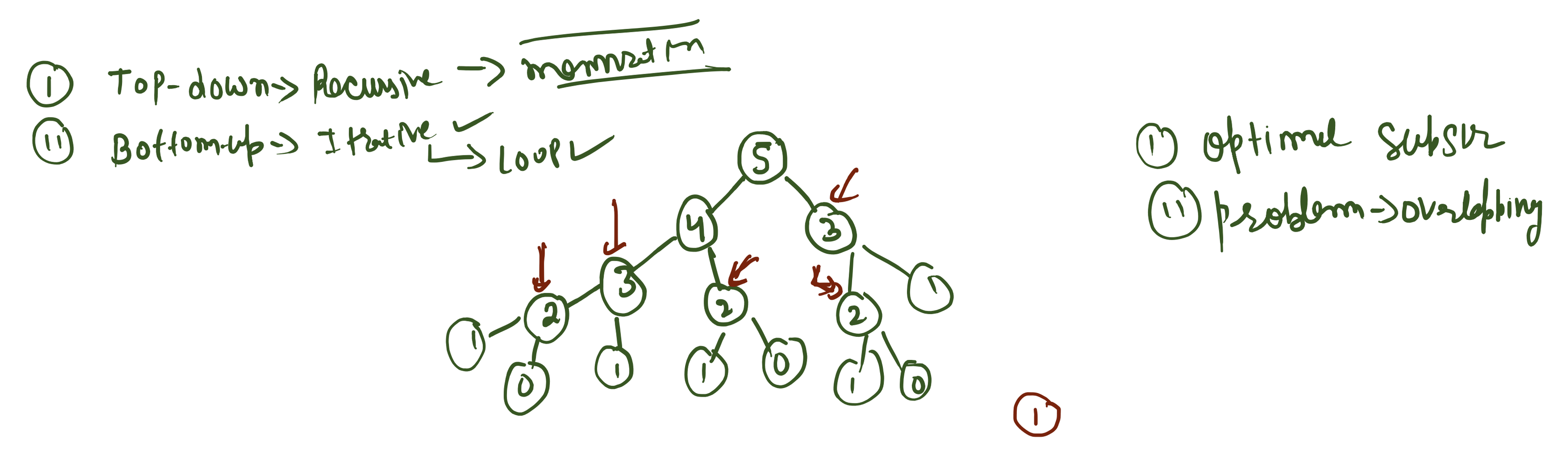
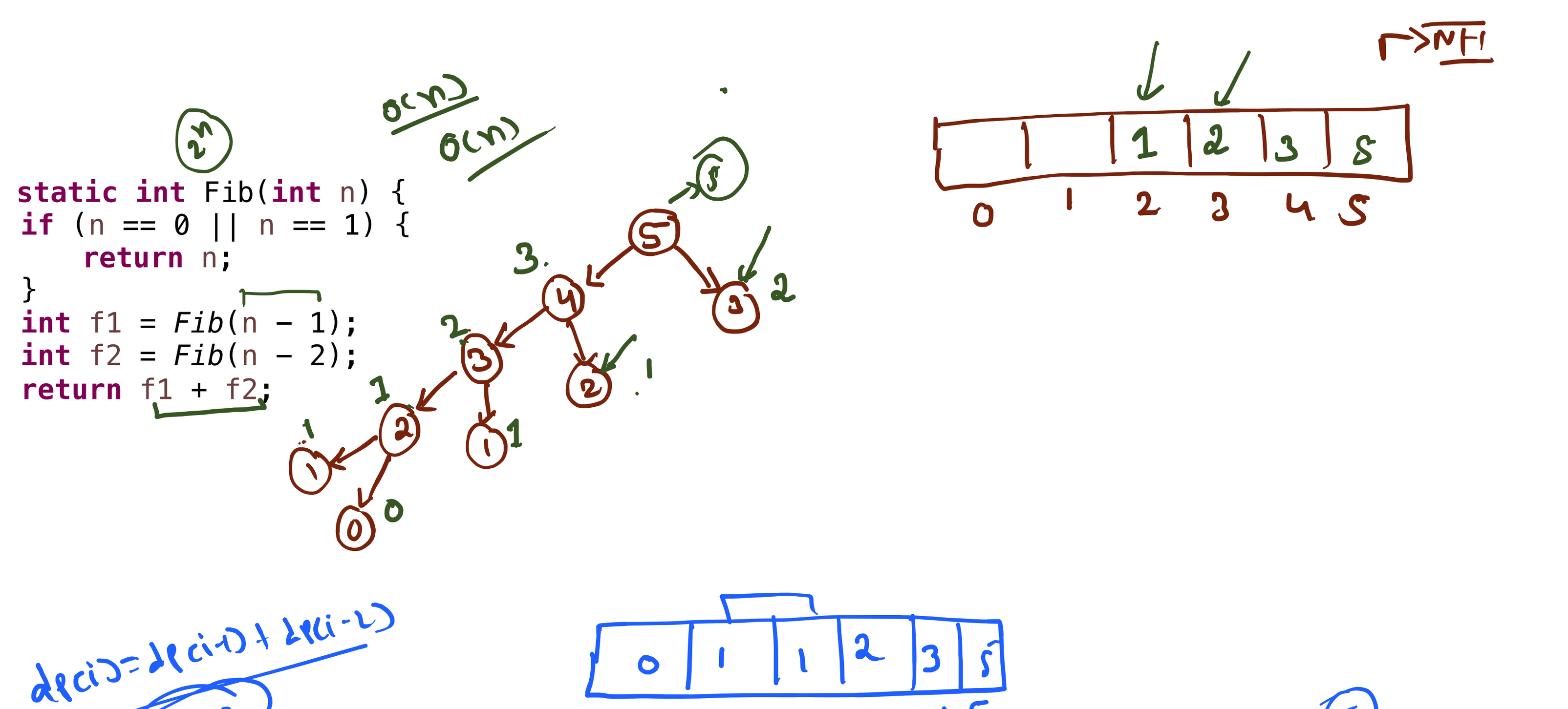
Lec-40

25/07/26 10:06 AM

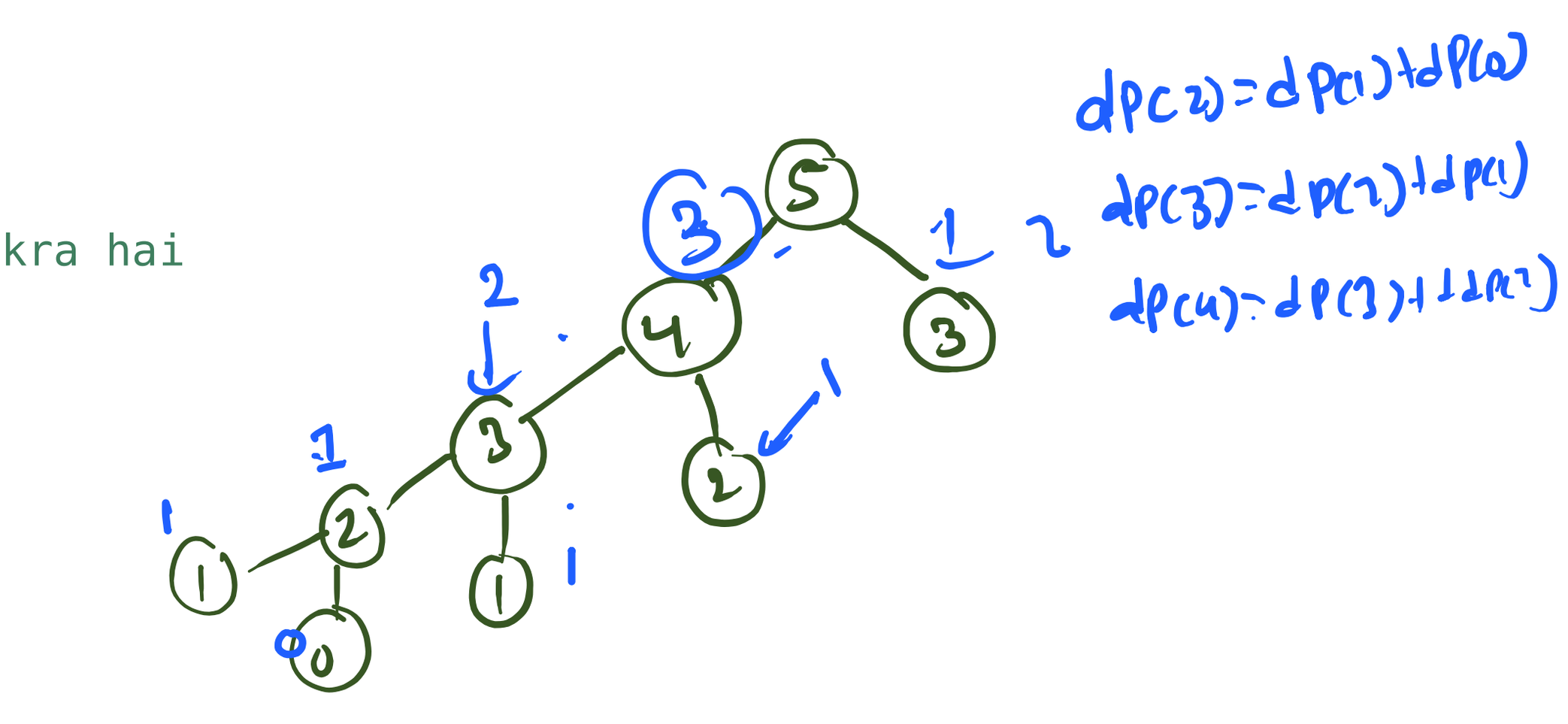


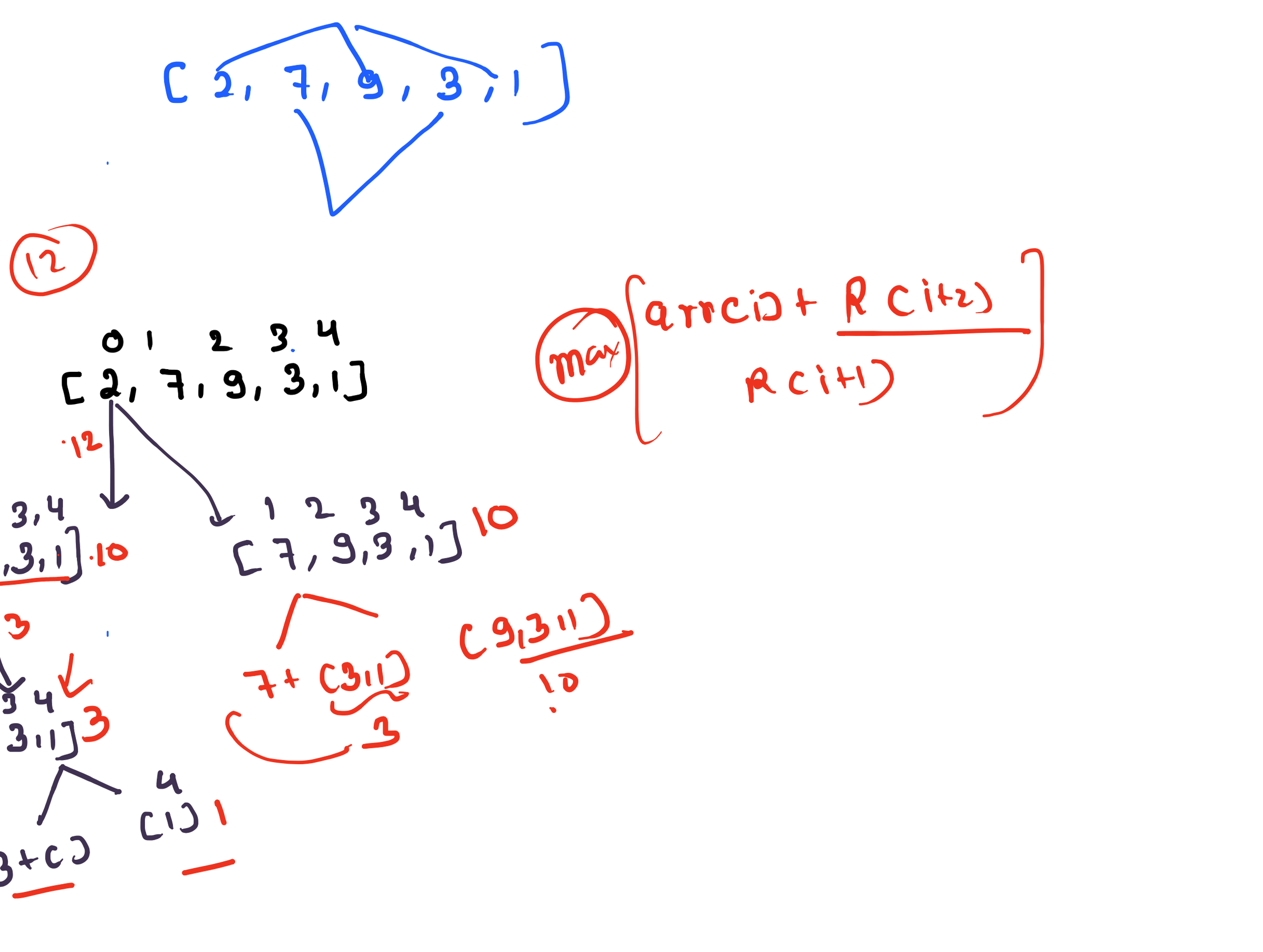


```
public static int Fib(int n) {
        if (n == 0 || n == 1) {
            return n;
        }
        int f1 = Fib(n - 1);
        int f2 = Fib(n - 2);
        return f1 + f2;

    }
```

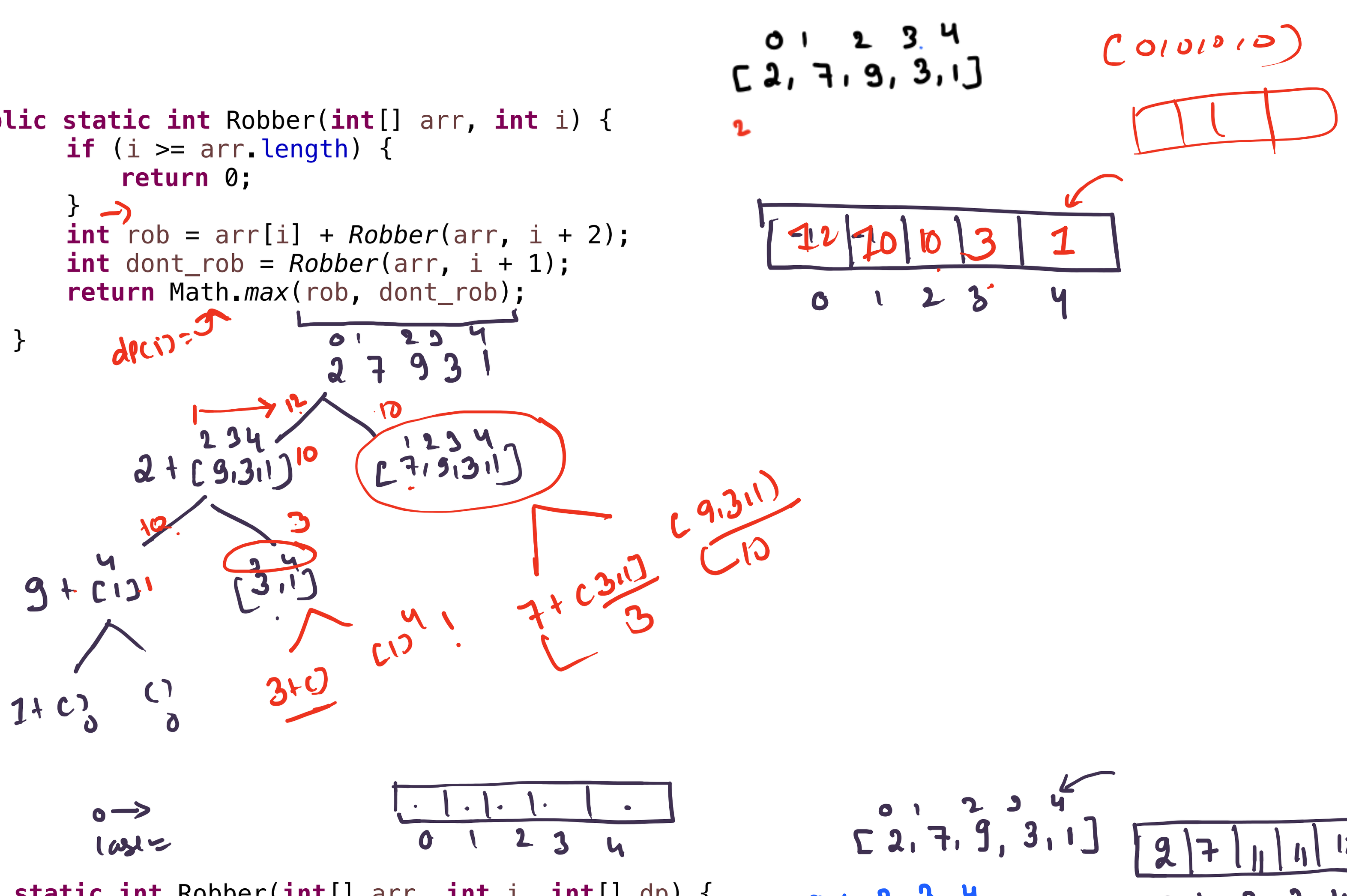


```
public static int FibTD(int n, int[] dp) {
        if (n == 0 || n == 1) {
            return n;
        }
        if(dp[n]!=0) {
            return dp[n];
        }
        int f1 = FibTD(n - 1, dp);
        int f2 = FibTD(n - 2, dp);
        return dp[n]=f1 + f2;// yaad kra hai

    }
```

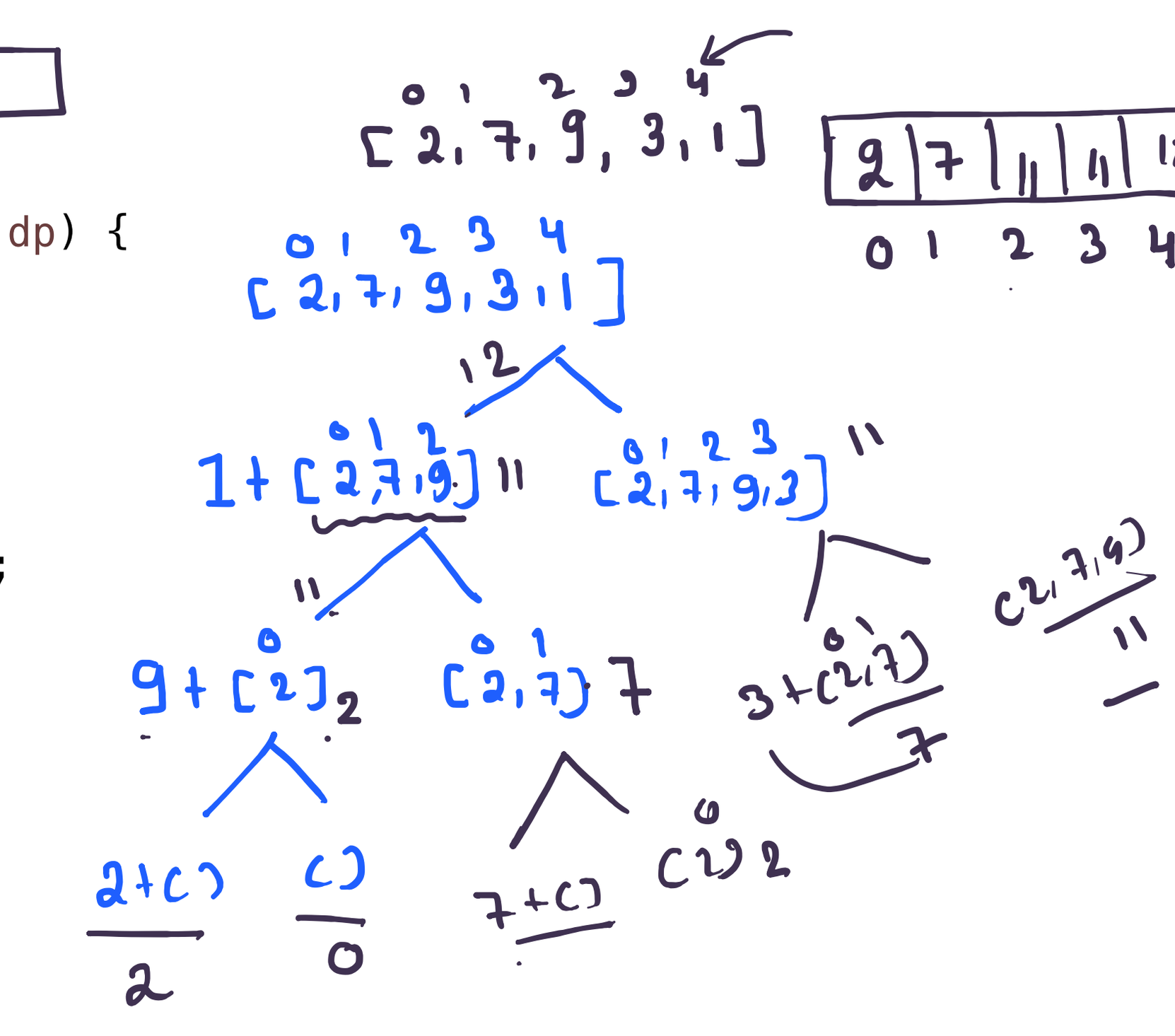


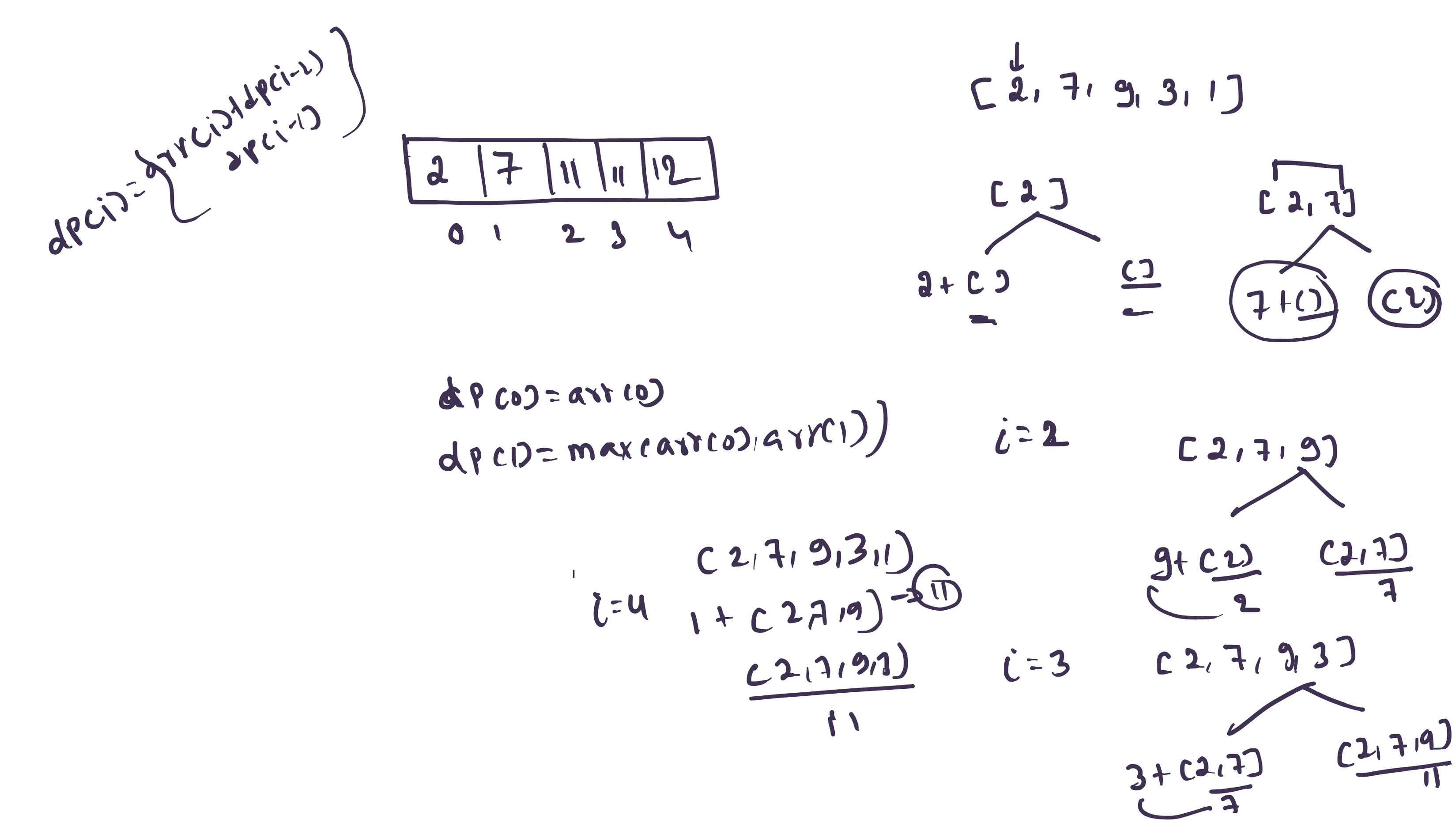


```
public static int Robber(int[] arr, int i) {
        if (i >= arr.length) {
            return 0;
        }
        int rob = arr[i] + Robber(arr, i + 2);
        int dont_rob = Robber(arr, i + 1);
        return Math.max(rob, dont_rob);

    }
```



```
public static int Robber(int[] arr, int i, int[] dp) {
        if (i >= arr.length) {
            return 0;
        }
        if (dp[i] != -1) {
            return dp[i];
        }
        int rob = arr[i] + Robber(arr, i + 2, dp);
        int dont_rob = Robber(arr, i + 1, dp);
        return dp[i] = Math.max(rob, dont_rob);

    }
```





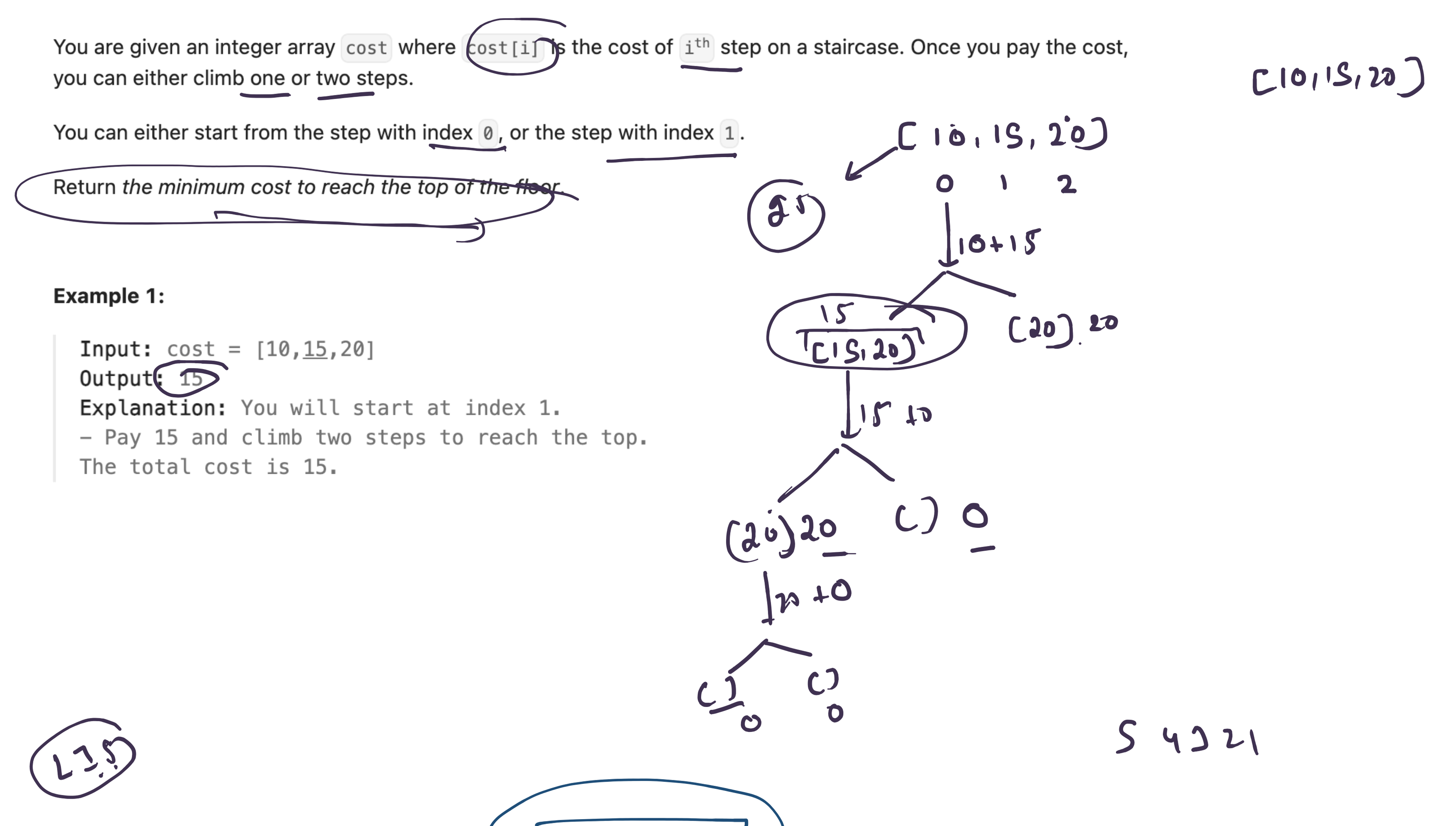
You are given an integer array `cost` where `cost[i]` is the cost of `ith` step on a staircase. Once you pay the cost, you can either climb one or two steps.

You can either start from the step with index `0`, or the step with index `1`.

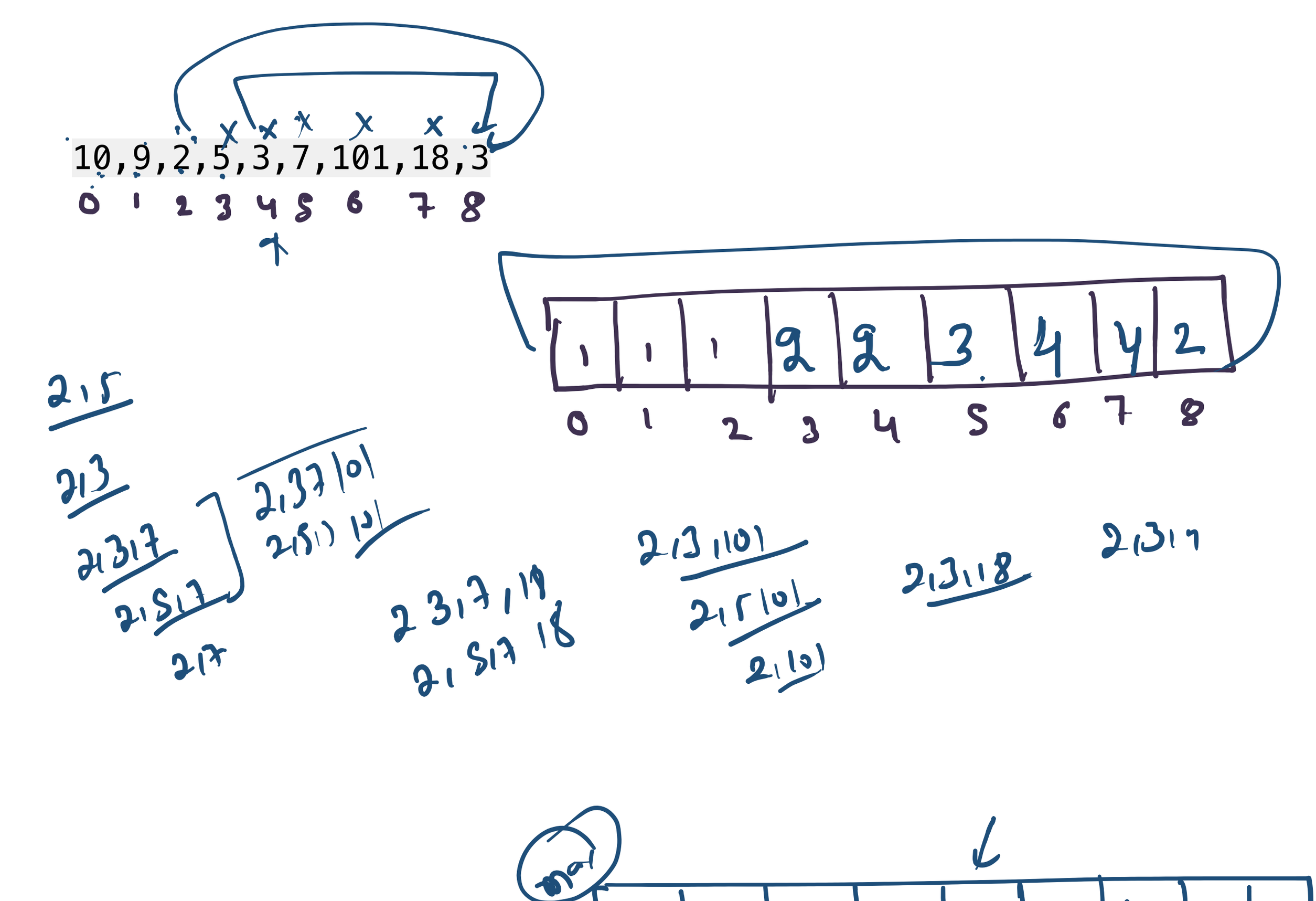
Return *the minimum cost to reach the top of the floor.*

Example 1:

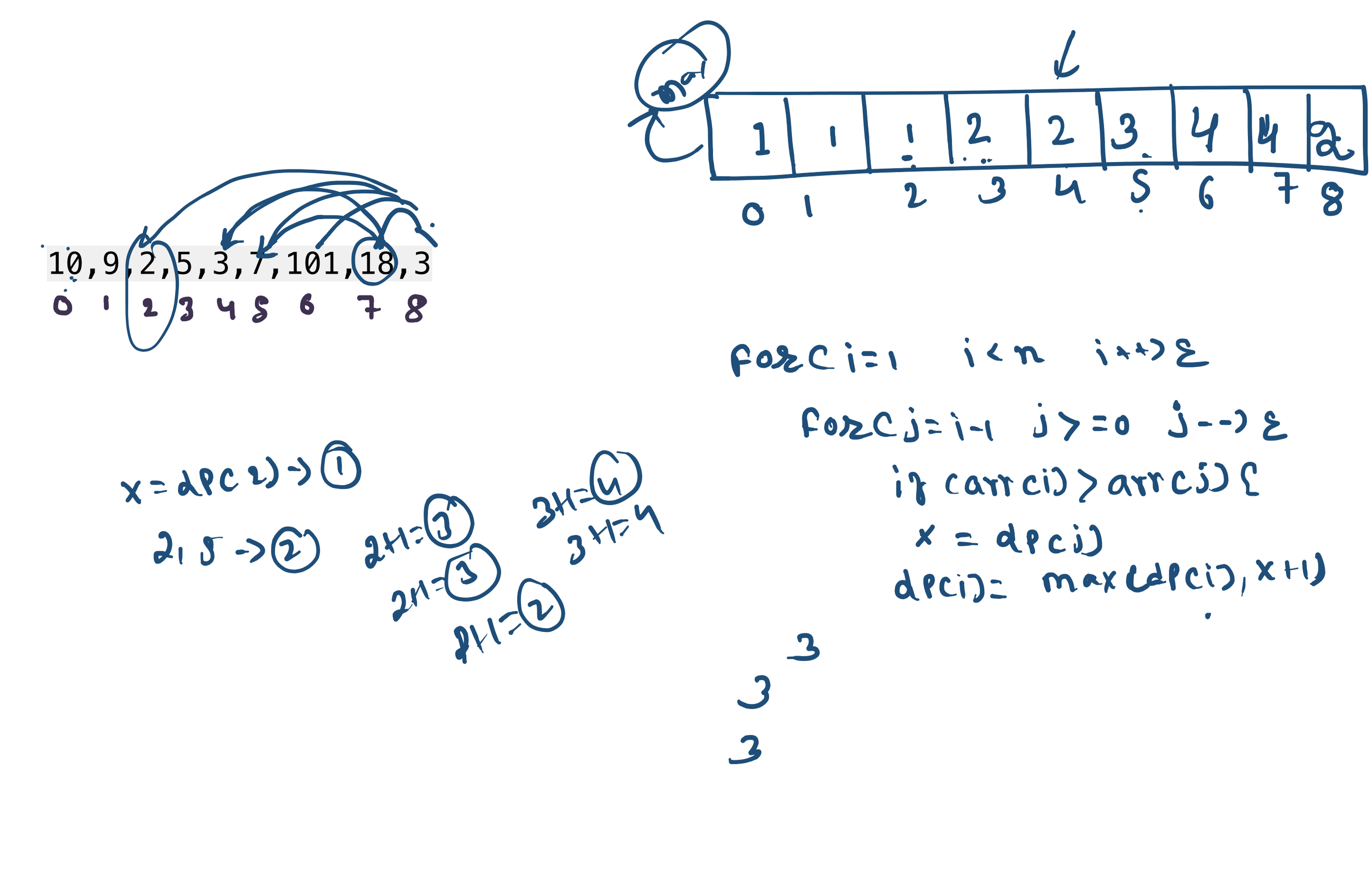
```
Input: cost = [10,15,20]
Output: 15
Explanation: You will start at index 1.
- Pay 15 and climb two steps to reach the top.
The total cost is 15.
```

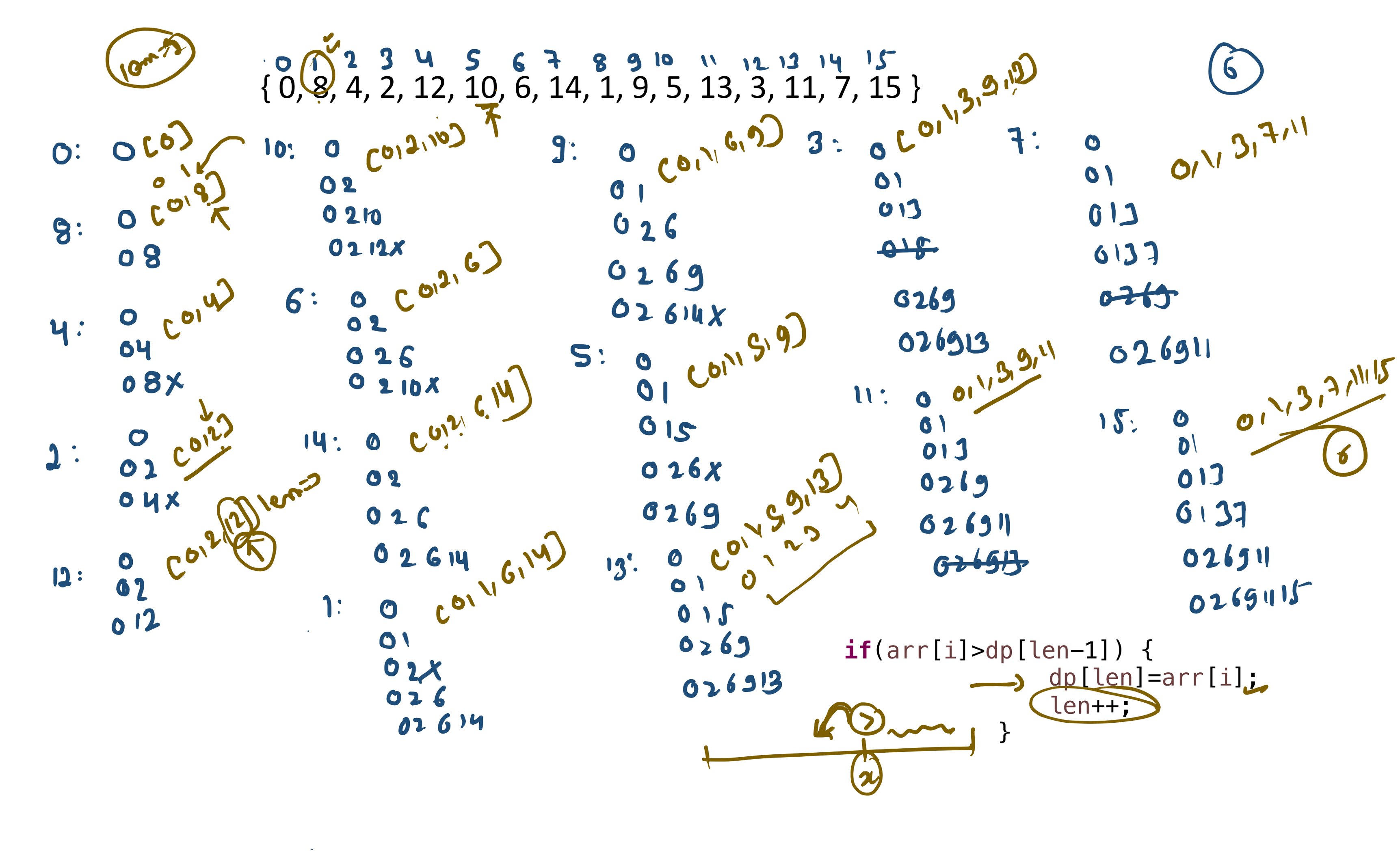


10,9,2,5,3,7,101,18,3



10,9,2,5,3,7,101,18,3





```
if(arr[i]>dp[len-1]) {
        dp[len]=arr[i];
        len++;
    }
```